

# NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

**Exam:** UPSC CDS

**Subject:** Mathematics

**Questions:** 208

**Source:** Official source PDF extraction

**AI-generated PYQs:** 0

## 1. [UNKNOWN] unknown

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

## 2. [UNKNOWN] unknown

1 - 2 sin<sup>0</sup> (0 2 cos<sup>0</sup> + 1 (d) 1 - 2 cos<sup>0</sup> SPSS-T-XHI ( 1 6 - A ) ===== PAGE 17 =====

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## 3. [UNKNOWN] unknown

1 - 2 cos<sup>0</sup> - cos ( 1 7 - A ) ===== PAGE 18 =====

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## 4. [UNKNOWN] unknown

1. (sec<sup>2</sup> 0 - 1) (1 - cosec<sup>2</sup> 0) = 1 (0 H - 3h units

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## 5. [UNKNOWN] unknown

1. (sec<sup>0</sup> - 1) (1 - cosec<sup>0</sup>) = 1 (d) H - 4h units

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## 6. [UNKNOWN] unknown

1. The orthocentre of a triangle always lies inside the triangle.

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## 7. [UNKNOWN] unknown

1. Three angles of one triangle are equal to the corresponding three angles of the other triangle.

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## 8. [UNKNOWN] unknown

1. 3iRT<sup>^</sup>% "5<sup>^</sup>^T5[, S T P T<sup>^</sup> T g r T f F T 3T<sup>^</sup>^ »+)<li ' Z T T 3 T S<sup>^</sup> c J ( S r r f<sup>^</sup>^ I %, ^ «i<^H I

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## 9. [UNKNOWN] unknown

2. Please note that it is the candidate's responsibility to encode and fill in the Roll Number and Test Booklet Series Code A, B, C or D carefully and without any omission or discrepancy at the appropriate places in the OMR Answer Sheet. Any omission/discrepancy will render the Answer Sheet liable for rejection.

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## 10. [UNKNOWN] unknown

2. t ? a + b = 2c|, ^ ^TTH cH^I a - c b - c (a) - 1 (b) 0 (c) 1 (d) 2

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**11. [UNKNOWN] unknown**

2.  $\sin \theta (1 + \cos \theta)^n + (1 + \cos \theta) (\sin \theta)^n = (d) H - 4h \text{ units } 2 \csc \theta$  Which of the above is/are correct ?

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**12. [UNKNOWN] unknown**

2.  $\sin \theta (1 + \cos \theta)^n + (1 + \cos \theta) (\sin \theta)^n = A^n t / f ? 2 \csc \theta 60. \sec X \csc x t ?$  (a)  $^{\text{cid } 1}$  (a) 2 (b)  $^{\text{cid } 2}$  (b)  $2n+1$  (c)  $13^2 < H$  (c)  $2^n$  (d)  $^{\text{cid } 1} 3^2$  (d) 211-1

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**13. [UNKNOWN] unknown**

2 -  $\sqrt{3}$  (b)  $\sqrt{3}$  - i (c)  $2 + \sqrt{3}$  (d)  $\sqrt{3} + 1$  SPSS-T-XHI ( 2 0 - A)

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**14. [UNKNOWN] unknown**

2 - S t ? (b)  $\sqrt{3}$  - i (a) 5 :3 (c)  $2 + S$  (b)

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**15. [UNKNOWN] unknown**

2. The centroid of a triangle always lies inside the triangle.

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**16. [UNKNOWN] unknown**

2. Three sides of one triangle are equal to the corresponding three sides of the other triangle,

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**17. [UNKNOWN] unknown**

2.  $T^{\text{cid } 7} \# T^{\text{cid } 3} T [ \% \# I$

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**18. [UNKNOWN] unknown**

2 : S n (c)  $2(a + b + c) > 3(p + q + r)$  (d)

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**19. [UNKNOWN] unknown**

2.  $AB \times EF = BC \times DE$  Which of the above is/are correct ? (a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2 In the equilateral triangle ABC given below,  $AD = DB$  and  $AE = EC$ . If  $l$  is the length of a side of the triangle, then what is the area of the shaded region ? A (a) (b) (0 (d)  $373 \frac{16}{32} 3^{\frac{16}{32}} 32 \frac{3}{2} 32$

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**20. [UNKNOWN] unknown**

2.  $AB \times EF = BC \times DE$   $t^{\text{cid } 1} / f ?$  (a) 1 (b) 2 (c)  $13^2$  (d)  $^{\text{cid } 1} c f t l, q^2$  ABC  $AD = DB$   $3^{\text{cid } 1} AE = EC$   $t^{\text{cid } 1} -^{\text{cid } 1} T i^{\text{cid } 1} t, ^{\text{cid } 1} T^{\text{cid } 1} r t f^{\text{cid } 1} d^{\text{cid } 1} W t ?$  A (a) (b) (c) (d)  $3\sqrt{3} f 16 \frac{32}{3} 3^{\frac{16}{32}} 32 \frac{3}{2} 32 3^{\frac{16}{32}}$

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**21. [UNKNOWN] unknown**

2.  $^{\text{cid } 1} W S 5 T F T \% O M R i^{\text{cid } 1}, i f^{\text{cid } 1} ^{\text{cid } 1} y 3^{\text{cid } 1} J^{\text{cid } 1} W c h l a T^{\text{cid } 1} p T^{\text{cid } 1} ^{\text{cid } 1})$  A, B, C  $^{\text{cid } 1} D S 2 T R^{\text{cid } 1} T^{\text{cid } 1} ^{\text{cid } 1} ^{\text{cid } 1} S t l^{\text{cid } 1} ^{\text{cid } 1} j j c n K^{\text{cid } 1} ^{\text{cid } 1} " S T R - M J i c h f^{\text{cid } 1} i j j N J i i i^{\text{cid } 1} ^{\text{cid } 1} I$

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**22. [UNKNOWN] unknown**

3. You have to enter your Roll Number on the Test Booklet in the Box provided alongside. DO NOT write anything else

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3.  $t^{3TN} t^{37W} | J_{ifdch} | t^{3W} |$

**34. [UNKNOWN] unknown**

4. This Test Booklet contains 100 items (questions). Each item is printed both in Hindi and English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

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**35. [UNKNOWN] unknown**

4. (a) 7 (b) 9 (c) 11 (d) 13 If  $a + b = 2c$ , then what is the value of  $a - c$  b - c (a) - 1 (b) 0 (c) 1 (d) 2 If  $X = y =$  and  $z =$  where  $x \perp 1, y \perp l, z \perp l$ , then what is the value of  $abc$  ? (a) - 1 (b) 1 (a) 2 (b) 3 (0 0 (c) - 2 (d) 3 (d) - 3 If  $2b = a + c$  and  $y^2 = xz$ , then what is 8. If  $y \propto x^2$  -  $c \propto C - a - b$  equal to ?  $ab + xy$   $bx$  (a) 3 (a) 1 (b) 2 (b) 2 (c) 1 (c) 0 (d) - 1

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**36. [UNKNOWN] unknown**

4.  $\angle m_1 < 2b = a + c$   $3 \perp y^2 = xz$   $f, \perp xb - CyC - azB - b$  (a) 3 (b) 2 (c) 1 (d) - 1 SPSS-T-XHI (3

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**37. [UNKNOWN] unknown**

4. The centroid of a right angled triangle lies on the triangle. Which of the above statements are correct ? (a) 1 and 2 (b) 1 and 4 (0 2 and 3 (d) 2 and 4 The locus of a point equidistant from two intersecting lines is (a) A straight line (b) A circle (c) A pair of straight lines (d) None of the above

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**38. [UNKNOWN] unknown**

4. Two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangle. Which of the above statements are correct ? (a) 1, 2 and 3 (b) 1, 3 and 4 (c) 1, 2 and 4 (d) 2, 3 and 4

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**39. [UNKNOWN] unknown**

4. " $Q \perp \angle hh4 \perp J \perp \perp \perp f$  ? (a)  $1 \perp 3 \perp 2$  (b)  $1 \perp 3 \perp 4$  (c)  $2 \perp 3 \perp 3$  (d)  $2 \perp 3 \perp 4$  '

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**40. [UNKNOWN] unknown**

4.  $\perp \blacksquare \perp STT \perp T \perp \% 3 \perp 7 \perp \perp \perp \% \# i c f cl8n 3 \perp 3 \perp \perp \perp \perp \perp f$  ? (a) 1,2 3fk3 (b) 1,3 3  $\perp 4$  (c) 1,2 3  $\perp 4$  (d) 2, 3 3  $\perp 4$

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**41. [UNKNOWN] unknown**

4 : 3 k (a) (b) (c) (d)  $27ir \perp - j47T \perp + 9 \text{ TOC}' + 1 \text{ 7ur}$

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**42. [UNKNOWN] unknown**

4 : 371 (d) 3 (a + b + c) > 4 (p + q + r) SPSS-T-XHI (27 - A )

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**43. [UNKNOWN] unknown**

4 . W JifWchl  $\perp 1 \perp 0 \perp 0$  (t o )  $\perp f \perp l \perp 3 \perp T \perp 3 \perp 1 \perp ? \perp 3 \perp 3 \perp T \perp \perp \perp \perp 1 \perp 5 \perp t \perp l \perp 3 \perp [r \perp \perp \perp 3 \perp R \perp \perp (:3 \text{ m } ) \perp \perp \perp f \perp l \perp TT \perp \perp \perp l c \perp \perp tR \perp \perp \perp 3 \perp TFT \perp f \perp l \perp 3 \perp T \perp T \perp W \perp \perp \perp 3 \perp T \perp f \perp \perp \perp frgrTT \perp f \perp , \perp \perp \perp 3 \perp 1 \perp ? \perp \perp \perp 3 \perp ? \perp f \perp \perp \perp \# 3 \perp TTWt \perp \perp l \perp 3 \perp l c \% FJ \perp 3 \perp R \perp 5 \perp r \perp R \perp \perp 5 \perp t \perp l$

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**44. [UNKNOWN] unknown**

5. You have to mark all your responses ONLY on the separate Answer Sheet provided. See directions in the Answer Sheet.

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**45. [UNKNOWN] unknown**

5.  $37^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot 1^{\frac{1}{3}}$  tht  $^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot f \mid ^{\frac{1}{3}} \cdot it \mid$

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**46. [UNKNOWN] unknown**

7. Before you proceed to mark in the Answer Sheet the response to various items in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per instructions sent to you with your Admission Certificate.

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**47. [UNKNOWN] unknown**

7. Which one of the following is correct ? (a) Decimal expansion of a rational number is terminating. (b) Decimal expansion of a rational number is non-terminating. (c) Decimal expansion of an irrational number is terminating. (d) Decimal expansion of an irrational number is non-terminating and non-repeating. If the roots of the equation  $px^2 + x + r = 0$  are reciprocal to each other, then which one of the following is correct ? (a)  $P = 2r$  (b)  $P = r$  (c)  $2p = r$  (d)  $P = 4r$  If  $65x - 33y$  what is  $xy$  equal to ? equal to ? (d) - 1 SPSS-T-XHI ( 2 - A ) ===== PAGE 3 =====  $517 + 5^8 + 5^9 + 5^{10}$  t ? (a) 7 (b) 9 (c) 11 (d) 13

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**48. [UNKNOWN] unknown**

7.  $^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot N^{\frac{1}{3}} \cdot W^{\frac{1}{3}} + \mid \% t o M \% 3^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot u \cdot T^{\frac{1}{3}} - ^{\frac{1}{3}} \% ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \% 3^{\frac{1}{3}} \cdot m^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot f \cdot i$

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**49. [UNKNOWN] unknown**

8. After you have completed filling in all your responses on the Answer Sheet and the examination has concluded, you should hand over to the Invigilator only the Answer Sheet. You are permitted to take away with you the Test Booklet.

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**50. [UNKNOWN] unknown**

8.  $Pn id R^{\frac{1}{3}} \cdot d^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot t^{\frac{1}{3}} ?$  (a)  $^{\frac{1}{3}} \cdot c^{\frac{1}{3}} \cdot H^{\frac{1}{3}} \cdot d^{\frac{1}{3}} \cdot c^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot \# \cdot c^{\frac{1}{3}} \cdot h^{\frac{1}{3}} \cdot t^{\frac{1}{3}} \mid$  (b)  $^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot F^{\frac{1}{3}} \cdot t^{\frac{1}{3}}$  (c)  $3^{\frac{1}{3}} \cdot n^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot c^{\frac{1}{3}} \cdot (i^{\frac{1}{3}} \cdot m^{\frac{1}{3}} \cdot d^{\frac{1}{3}} \cdot c^{\frac{1}{3}} \mid 3^{\frac{1}{3}} \cdot r^{\frac{1}{3}} \cdot a^{\frac{1}{3}} \cdot K^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot d^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot F^{\frac{1}{3}} \cdot t^{\frac{1}{3}}$  (d)  $^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot 7^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot \# \cdot c^{\frac{1}{3}} \cdot n^{\frac{1}{3}} \cdot t^{\frac{1}{3}} \mid ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot i^{\frac{1}{3}} \cdot f^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot U^{\frac{1}{3}} \cdot T^{\frac{1}{3}}$   $px^2 + x + r = 0$  %  $^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \% 05^{\frac{1}{3}} \cdot I^{\frac{1}{3}}, ^{\frac{1}{3}} \cdot P^{\frac{1}{3}} \cdot i^{\frac{1}{3}} \cdot H^{\frac{1}{3}} \cdot f^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot k^{\frac{1}{3}} \cdot d^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot t^{\frac{1}{3}} ?$  (a)  $p = 2r$  (b)  $p = r$  (c)  $2p = r$  (d)  $p = 4r$  If  $65x - 33y = 97$   $3^{\frac{1}{3}} \cdot 33x - 65y = 1$  t ,  $^{\frac{1}{3}} \cdot xy \% ? T \% WJ^{\frac{1}{3}} \cdot t^{\frac{1}{3}} ?$  (a) 2 (b) 3 (c) 0 - 2 (d) - 3  $z, ^{\frac{1}{3}} \cdot a^{\frac{1}{3}} \cdot b^{\frac{1}{3}} + s^{\frac{1}{3}} \cdot y^{\frac{1}{3}} \cdot c^{\frac{1}{3}} \cdot z^{\frac{1}{3}} \cdot a^{\frac{1}{3}} \cdot b^{\frac{1}{3}} \cdot t^{\frac{1}{3}} ?$  (a) 1 (b) 2 (c) 0 (d) - 1 ===== PAGE 4 =====

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**51. [UNKNOWN] unknown**

8.  $3^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot N^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot r^{\frac{1}{3}} \cdot I^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot J^{\frac{1}{3}} \cdot f^{\frac{1}{3}} \cdot R^{\frac{1}{3}} \cdot W^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot I^{\frac{1}{3}} \mid 3^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot N^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot m^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot f^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot ^{\frac{1}{3}} \cdot 3^{\frac{1}{3}} \cdot T^{\frac{1}{3}} \cdot i^{\frac{1}{3}} \cdot r^{\frac{1}{3}} \cdot f^{\frac{1}{3}} \cdot t^{\frac{1}{3}} \mid$

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**52. [UNKNOWN] unknown**

9. Sheets for rough work are appended in the Test Booklet at the end.

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**53. [UNKNOWN] unknown**

9. If  $a^{\frac{1}{3}} - 1 = 5$ , then what is the value of  $a^{\frac{1}{3}} - 1$  (a) 125 (b) -125 (c) 140 (d) -140

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**54. [UNKNOWN] unknown**

9.  $i = 5t$ ,  $t$  TH W a (a) 125 (b) - 125 (c) 140 (d) - 140

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**55. [UNKNOWN] unknown**

10. Penalty for wrong answers : THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS. (i) There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty. (ii) If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question. (iii) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

\_\_\_\_\_ DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE TOLD TO DO SO \_\_\_\_\_ SPSS-T-XHI  
(1 - A) ===== PAGE 2 ===== q17 j\_ gi8 ^ 5I9 ^ g20 -g divisible by

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**56. [UNKNOWN] unknown**

10. If  $X + y + z = 0$ , then what is  $(y + z - x)^4 + (z - x - y)^4 + (x + y - z)^4$  equal to? (a)  $(x + y + z)^4$  (b)  $3(x + y)(y + z)(z + x)$  (c)  $24xyz$  (d)  $-24xyz$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**57. [UNKNOWN] unknown**

10.  $x + y + z = 0$ ,  $(y + z - x)^4 + (z + x - y)^4 + (x + y - z)^4$  ter% t ? (a)  $(x + y + z)^4$  (b)  $3(x + y)(y + z)(z + x)$  (c)  $24xyz$  (d)  $-24xyz$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**58. [UNKNOWN] unknown**

10. <icnd v3TiO %f^TTJ c|TJ5 ; <iufi<^e4H^ f ^ T T T T T | H H 3 ^ % f ^ ^ f ^ ^ i m ^ I (i) 5 R ^ T O % f ^ ^ 3 ^ f I 3 C T fl ^ ^ 3 I r ^ T O % %Xi 1 ^ ^ T T ^ 3 ^ % f ^ T O ^ 1%IT Tfii 37^ ^ ^ ^ ^ ^ nxJT T T I (ii) ^ ^ 3 o f t ^ T T ^ 3 T te i 3 m ^ t , rft ^ 3 tR ^ TRT ^ 5ni?7n, 1 ^ Wl, 3rlfr ^ T ^ TJcR ir R ^ ^ 1 , 1 ^ ^ 3 H T O % f ^ 3M^xby^K ^ 3 ^ ^ ^ f ^ ^ mjU I (iii) ^ 3 C T ft ^ ^ ^ T O ^ W T t , 3 T ^ 3ufl^ciK ^ 3 m ^ f ^ W T ^ 3H T O % ^ rnnn i 37^ ?7^ Sifwcf)) 27^ ^ if^flTr^f^chi isiln^ ^ W 377^ 17^ ?7^ W <Sfl(^ Note : English version of the instructions is printed on the front cover of this Booklet. SPSS-T-XHI ( 36 - A )

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**59. [UNKNOWN] unknown**

11. If  $(x + 3)$  is a factor of  $x^4 + 3x^3 + 4x^2 + kx$ , then what is the value of  $k$  ?

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**60. [UNKNOWN] unknown**

11.  $(x + 3)$ ,  $x^4 + 3x^3 + 4x^2 + kx + t$ ,  $^n qn w t$  ? (a) 12 (b) 24 (c) 36 (d) 72

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**61. [UNKNOWN] unknown**

12. The smallest integer with 4 digits which is a perfect square is (a) 1000 (b) 1024 (c) 1089 (d) None of the above  
SPSS-T-XHI ( 4 - A)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**62. [UNKNOWN] unknown**

12.  $4 \cdot 37^4 \cdot T J c R$ ,  $f n M i d H i^d$ ,  $^n qn w t$  ? (a) 1000 (b) 1024 (c) 1089 (d) ^ ^ ^

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**63. [UNKNOWN] unknown**

13. Which one of the following is a zero of the polynomial  $3x^4 + 4x^3 - 7$  ? (a) 0 (b) 1 (c) 2 (d) - 1

64. [UNKNOWN] unknown

13.  $3x^3 + 4x^2 - 7x + 1$  at  $x = 2$ ? (a) 0 (b) 1 (c) 2 (d) -1

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf)

65. [UNKNOWN] unknown

14. There are two numbers which are greater than 21 and their LCM and HCF are 3003 and 21 respectively. What is the sum of these numbers ? (a) 504 / (b) 508 (c) 514 (d) 528 (a) 12 (b) 24

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf)

66. [UNKNOWN] unknown

14.  $f^{211} f^3 \wedge \wedge \text{LCM } 3 \wedge \text{HCF } 3003 \wedge 21 f \wedge \wedge T^3 \text{Tr}^4 > l^+$  d W t ? (a) 504 (b) 508 (c) 514 (d) 528

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf)

**67. [UNKNOWN] unknown**

15. If  $a$  and  $p$  are the roots of the equation  $0.36 \cdot ax^2 + bx + c = 0$ , then what is the value of the expression  $(a + 1)(3 + 1)$  ? (a) (b) (c) (d)  $a + b + c$   $a + b - c$   $a - b + c$   $a + b - c$  ===== PAGE 5 =====

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf)

68. [UNKNOWN] unknown

15.  $a^3(3^4) \ll h \ll u \mid ax^h + bx + c = 0 \pmod{f}, (a+1)(p+1) \nmid \text{TO Wt?}$  (a) (b) (0) (d)  $a+b+c \mid b+c-a \mid a-b+c \mid a+b-c$  SPSS-T-XHI (5-A) ===== PAGE 6 =====

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf)

**69. [UNKNOWN] unknown**

15- 6 2 C N - 4 A B 60 (a) 10,- 3 0 ,-1 (b) 10.- 1, 30 (c) -30, 1 0,-1 (d) - 1 , -- 30, 10 A person borrowed ^ 5,000 at 5% rate of interest per annum and immediately lent it at 5-5%. After two years he collected the amount and settled his loan. What is the amount gained by him in this transaction ? (a) ^ 25 (b) ^ 50 (c) ^ 100 (d) ^ 200 SPSS-T-XHI ( 8 - A ) =====  
PAGE 9 =====

Official source: [backend/data/pvg/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf](https://backend/data/pvg/collector/universal-official-pdfs/raw/OP-CDSI18-Math.pdf)

70. [UNKNOWN] unknown

15 - 6 2 C N - 4 A B 60 (a) (b) (c) (d) 10, - 3 0 , - 1 10, - 1, 30 - 30, 10, - 1 - 1, - 30, 10 (a) 75 (b) 85

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf)

71. [UNKNOWN] unknown

16. The remainder when  $3x^4 + kx^3 + 5x - 6$  is divided by  $(x + 1)$  is  $-7$ . What is the value of  $k$ ? (a)  $-14$  (b)  $14$  (c)  $-7$  (d)  $7$

If  $f(x)$  and  $g(x)$  are polynomials of degree  $p$  and  $q$  respectively, then the degree  $\{f(x) \pm g(x)\}$  (if it is non-zero) is (a) Greater than  $\min(p, q)$  (b) Greater than  $\max(p, q)$  (c) Less than or equal to  $\max(p, q)$  (d) Equal to  $\min(p, q)$

What is the value of  $\frac{1}{\sqrt{5} + \sqrt{3}} + \frac{1}{\sqrt{5} - \sqrt{3}}$ ? (a)  $-2\sqrt{15}$  (b)  $2\sqrt{15}$  (c)  $\sqrt{15}$  (d) What is the value of  $\frac{1}{1 + x} + \frac{1}{1 + x^2} + \frac{1}{1 + x^4} + \dots$

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72. [UNKNOWN] unknown

16.  $3x^3 + kx^2 + 5x - 6$  is a factor of  $(x + 1)^7$ . Find the value of  $k$ . (a) -14 (b) 14 (c) -7 (d) 7

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf)

73. [UNKNOWN] unknown

17.  $\wedge \text{fix } 3 \wedge g(x) \text{ fM } p \ 3 \wedge q \ \% \text{ cfr } \{\text{fix} \mid \pm g(x)\} \wedge l \ 1M \mid (a) \min(p, q) \wedge (b) \max(p, q) \wedge (c) \max(p, q) \% W \ m \wedge \wedge (d) \min(p, q) \%$

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf)

**74. [UNKNOWN] unknown**

19.  $V_s + V_s - V_s - V_s$  (a)  $-2V$  (b)  $2V$  (c)  $V$  (d)  $-V$  a - b , „c — b (a) - 1 (b) 0 (0 1 (d) 3 SPSS-T-XHI

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**75. [UNKNOWN] unknown**

20.  $^{\wedge} \wedge 'M > l4 > d$  20 t l  $^{\wedge} \wedge$  t (a) - S 3T2RT 4 (b) 2 3 (c) - 5 (d)

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**76. [UNKNOWN] unknown**

21. a - c , - c l +  $x^{\wedge} - ^{\wedge}$  + X where x 0 ? (a) - 1 (b) 0 (c) 1 (d) 3 SPSS-T-XHI

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**77. [UNKNOWN] unknown**

21.  $^{\wedge} \wedge ( ^{\wedge} )$  25% t ,  $^{\wedge} 3T^{\wedge} 4 ^{\wedge} \wedge$  cTT $^{\wedge} \wedge \wedge$  ? (a) 1S% (b) 20% (c) 25% (d) 30%

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**78. [UNKNOWN] unknown**

22. The sum of a number and its square is 20. Then the number is (a) - 5 or 4 (b) 2 or 3 (c) - 5 only (d) 5 or - 4 If the price of wheat rises by 25%, then by how much percent must a man reduce his consumption in order to keep his budget the same as before ? (a) 15% (b) 20% (c) 25% (d) 30% of the students who registered did not 25 11

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**79. [UNKNOWN] unknown**

22. % %xi;  $^{\wedge} 2S ^{\wedge} \wedge \wedge$  l  $^{\wedge} 20 ^{\wedge} \#$  f d  $^{\wedge} \wedge$  2000 t ,  $^{\wedge} 3rfh3f ^{\wedge} \wedge \wedge$  t ? (a) 1920 (b) 10S6 (c) 1020 (d) 864

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**80. [UNKNOWN] unknown**

23. appear for the examination, — of those who 20 appeared passed. If the number of registered students is 2000, the number who passed is (a) 1920 (b) 1056 (c) 1020 (d) 864 What is the difference between 0-9 and 0-9 ? (a) 0 (b) 0-099 (c) 0-1 (d) 0-09 ( 6 - A ) ===== PAGE 7 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**81. [UNKNOWN] unknown**

23. 0-9 3  $^{\wedge} 0-9$  % t ? (a) 0 (b) 0-099 (c) 0-1 (d) 0-09

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**82. [UNKNOWN] unknown**

24. If A : B = 1 : 2, B : C = 3 : 4, C : D = 2 : 3 and D : E = 3 : 4, then what is B : E equal to ? (a) 3 : 2 (b)

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**83. [UNKNOWN] unknown**

24.  $^{\wedge} A : B = 1 : 2$  ,  $B : C = 3 : 4$  ,  $C : D = 2 : 3$  3 fk D : E = 3 : 4 t ,  $^{\blacksquare} B : E$  f ? (a) 3 : 2 (b) 1 : 8 (c) 3 : 8 (d) 4-: 1

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**84. [UNKNOWN] unknown**

25. A work when done by 10 women is completed in 12 days. The same work can be completed in 8 days when done by 5 men. How many days will it take to complete when 6 women and 3 men are employed to perform the same job? (a) 12 (b) 10 (0 8 (d) 5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**85. [UNKNOWN] unknown**

26. A man undertakes to do a certain work in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. How many additional men should he employ so that the whole work is finished in time ? (a) 75 (b) 85 (c) 100 (d) 120

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**86. [UNKNOWN] unknown**

27. A train moving with a speed of 60 km per hour crosses an electric pole in 30 seconds. What is the length of the train in metres ? (a) 300 (b) 400 (c) 500 (d) 600

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**87. [UNKNOWN] unknown**

27.  $TT^{10} f^{12} W T I I 3^{5} 3TT^{f} S^{f} W T I^{1} \% TT^{6} 3 M 3^{3} 3TT^{WTR} f,^{1}$  ? (a) 12 (b) 10 (c) 8 (d) 5  $^{150} t I^{200} 3TT^{t} I W t \% 50 f^{35} FT^{1} W t I^{1} \%^{WTT} rf,^{1}$

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**88. [UNKNOWN] unknown**

28. A, B 3fk C  $^{120} W T^{f} A 20 3Tffe I 3fk C \% 3T^{2} 0^{t} I B^{3} 7^{W} |$  ? (a) 10 (b) 15 (c) 20 (d) 25

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**89. [UNKNOWN] unknown**

30.  $^{120}$  is distributed among A, B and C so that A's share is  $^{20}$  more than B's and  $W 20$  less than C's. What is B's share ? (a) 10 (b) 15 (c) 20 (d) 25 In the following table of inverse variation, what are the values of A, B and C respectively ? M

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**90. [UNKNOWN] unknown**

30.  $^{5\%}$  yid c[t[ s<sup>Tr3I</sup> | q r  $^{5,000}$  (c)  $^{100}$  sm  $^{<-d} ^{5-5\%} ^{1} ^{120} [ < u. 1 < 1 \ll K^{chl} f < ^{1} A i^{d4} 60$  km Mfd  $^{S} | ^{dm} ^{3TT} ? ^{30} ^{< h < t 1 \backslash i \backslash i} ^{dH} II, Tto ii W I$  ? (a)  $^{25}$  (a) 300 (b)  $^{50}$  (b) 400 (c) 500 (c)  $^{100}$  (d) 600 (d)  $^{200}$  SPSS-T-XHI ( 9 - A ) ===== PAGE 10 =====

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**91. [UNKNOWN] unknown**

31. At present the average of the ages of a father and a son is 25 years. After seven years the son will be 17 years old. What will be the age of the father after 10 years ? (a) 44 years (b) 45 years (c) 50 years (d) 52 years

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**92. [UNKNOWN] unknown**

31.  $25 I 10^{t} e^{37} T5 W \# ft$  ? (a)  $4^4$  (b)  $45^4$  (c)  $5^0$  (d)  $5^2$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**93. [UNKNOWN] unknown**

32. If 5 tractors can plough 5 hectares of land in 5 days, then what is the number of tractors required to plough 100 hectares in 50 days ? (a) 100 (b) 20 (c) 10 (d) 5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**94. [UNKNOWN] unknown**

32.  $^{5} pRX^{5} ^{5} ^{f}, 100^{50} \% cT$  (a) 100 (b) 20 (c) 10 (d) 5

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**95. [UNKNOWN] unknown**

33. A merchant commences with a certain capital and gains annually at the rate of 25%. At the end of 3 years he has

10,000. What is the original amount that the merchant invested ? (a) ? = 5,120 (b) ^ 5,210 (c) ^ 5,350 (d) ^ 5,500

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**96. [UNKNOWN] unknown**

33. oiJiMi^ TT^ Plf^d ^ t 3fk 25% ^ ^ t ^ W<T ^FJTcH t l 3 ^ ^r w r 10,000 ^ f i o^Nld % -gRT WTT^ nf 31Kte Ttllt W f ? (a) ^ 5,120 (b) r 5,210 (c) ^ 5,350 (d) ^ 5,500

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**97. [UNKNOWN] unknown**

34. Which one of the following decimal numbers is a rational number with denominator 37 ? (a) 0-459459459 . . . (a) 85 (b) 0-459459459 (b) 80 (c) 0-0459459459 . . . (c) 75 (d) 0-00459459 . . . (d) 70

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**98. [UNKNOWN] unknown**

34. Pnldl^d ^THd<^ ^ t ^ 1 ^ 371 ? (a) 0-459459459 . . . t ? (b) 0-459459459 (a) 85 (b) 80 (c) 0-0459459459 . . . (c) 75 (d) 0-00459459 . . . (d) 70

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**99. [UNKNOWN] unknown**

35. The annual income of a person decreases by T 64 if the annual rate of interest decreases from 4% to 3-75%. What is his original annual income ? (a) ^ 24,000 (b) r 25,000 (c) ^ 25,600 (d) ^ 24,600

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**100. [UNKNOWN] unknown**

35. ■2Tf^ ^ 4% ^ 3-75% ^ l , ^ X?:^ o?T% ^ 3TFT W 64 W. t 1 3TR W t ? (a) ' ? 24,000 (b) ^ 25,000 (c) r 25,600 (d) ^ 24,600

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**101. [UNKNOWN] unknown**

36. For  $0 < m < 1$ , which one of the following is correct ? (a) ■  $\log_j^Q m < m^\wedge < m < m^\wedge$  (b)  $m < m^\sim^\wedge < m^\wedge < \log_j Q m$  (c)  $\log_i n_i < m < m^\wedge < m^\wedge$  (d)  $\bullet \log_j^Q m < m^\wedge < m < m^\wedge$

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**102. [UNKNOWN] unknown**

36.  $0 < m < 1$  % f ^ , PIHfdfed ^ ^ ^ -F T ^ t ? (a)  $\log_i iii < n_i <$  (b)  $m < m^\wedge < m^\wedge < \log_j^Q m$  (c)  $\log_i n_i < m < m^\wedge < m^\wedge$  (d)  $\log_i ^\wedge < m < m^\wedge$

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**103. [UNKNOWN] unknown**

37. A gentleman left a sum of 39,000 to be distributed after his death among his widow, five sons and four daughters. If each son receives 3 times as much as a daughter receives, and each daughter receives twice as much as their mother receives, then what is the widow's share ? (a) ^ 1,000 (b) ^ 1,200 (c) 1,500 (d) None of the above

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**104. [UNKNOWN] unknown**

37. ^ ^ 39,000 ^ ^ ^ 3 ^ 3 ^ ^ ^ t l ^ ^ 3 ^ % ^ 3 3 ^ t , 3fk 31?^ 3 ^ Tff % f ^ ^pn f i f w t , ^ f|WT W t ? (a) ^ 1,000 (b) ^ 1,200 (c) ^ 1,500 (d) 3 q ^ ^ ^ ^ ^

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**105. [UNKNOWN] unknown**

38. Three numbers which are co-prime to each other, are such that the product of the first two is 286 and that of the last two is 770. What is the sum of the three numbers ? SPSS-T-XHI ( 1 0 - A ) ===== PAGE 11 =====

**106. [UNKNOWN] unknown**

38.  $31^{\circ}f, ^{\circ}W T t f^{\circ} 286 3^{\circ} 3Tf^{\circ} 770 t l^{\circ} W$  SPSS-T-XHI ( 11 - A ) ===== PAGE 12 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**107. [UNKNOWN] unknown**

39. The age of a woman is a two-digit integer. On reversing this integer, the new integer is the age of her husband who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages ? (a) 8 years (b) 9 years (c) 10 years (d) 11 years

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**108. [UNKNOWN] unknown**

39.  $3 W^{\circ} W^{\circ} f, ^{\circ} 3TT5^{\circ} W T l l 3TT5^{\circ} 3^{\circ}, 3^{\circ} 3TI5^{\circ} 41J| \gg + d^{\circ} J < MK^{\circ} c < T f | OT l l 3TT5^{\circ} 3T^{\circ} R W t ?$  (a)  $8^{\circ}$  (b)  $9^{\circ}$  (c)  $10^{\circ}$  (d)  $11^{\circ}$

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**109. [UNKNOWN] unknown**

40. A passenger train and a goods train are running in the same direction on parallel railway tracks. If the passenger train now takes three times as long to pass the goods train, as when they are running in opposite directions, then what is the ratio of the speed of the passenger train to that of the goods train ? (Assume that the trains nm at uniform speeds) (a) 2 :1 (b) 3 ; 2 (c) 4 :3 (d)

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**110. [UNKNOWN] unknown**

40.  $TT^{\circ} 3^{\circ} TTWnit^{\circ} WcR^{\circ} d^{\circ} Mit^{\circ} f l^{\circ} fe n^{\circ} cTt nr\#, TTiFrm^{\circ} cftq^{\circ} W T^{\circ} 37^{\circ} t l^{\circ} HTTH fe n^{\circ} T^{\circ} f, ci^{\circ} 373qr^{\circ} W t ?$  (TO cftf^{\circ} % ^irfM^{\circ} l) (a) 2 :1 (b) 3 :2 (c) 4 :3 (d) 1 :1

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**111. [UNKNOWN] unknown**

41. All odd prime numbers upto 110 are multiplied together. What is the unit digit in this product ? (a) 0 (b) 3 (c) 5 (d) None of the above

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**112. [UNKNOWN] unknown**

41.  $110 c!^{\circ} 1^{\circ} 3T^{\circ} 7T^{\circ} W n sff^{\circ} W T t l^{\circ} 3T^{\circ} l ?-$  (a) 0 (b) 3 (c) 5 (d) '3^{\circ}

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**113. [UNKNOWN] unknown**

42. An alloy A contains two elements, copper and tin in the ratio of 2 : 3, whereas an alloy B contains the same elements in the ratio of

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**114. [UNKNOWN] unknown**

42.  $f^{\circ} A^{\circ} r^{\circ} 3\#^{\circ} : -feq 2 : 3 \% 3?3qT^{\circ} f, ^{\circ} 51^{\circ} f^{\circ} 37^{\circ} B^{\circ} cf^{\circ} 3 : 4 \% 373qTcT^{\circ} f l^{\circ} 20 kg f^{\circ} m^{\circ} v^{\circ} A, 28 B, 5^{\circ} 15 3fiT 3 ? !^{\circ} dMt C t, f^{\circ} RT^{\circ} clkr 3^{\circ} 1^{\circ} 6 : 7 \% 373qm^{\circ} f, ^{\circ} C^{\circ}$  (a) 3 kg (b) 4 kg (c) 5 kg (d) 7 kg

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**115. [UNKNOWN] unknown**

43. A quadratic polynomial  $ax^2 + bx + c$  is such that when it is divided by  $x$ ,  $(x - 1)$  and  $(x + 1)$ , the remainders are 3, 6 and 4 respectively. What is the value of  $(a + b)$  ? (a) 3 (b) 2 (c) 1 (d) - 1

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**116. [UNKNOWN] unknown**

43. fencft ax<sup>2</sup> + bx + c ^ 3 ITO 1 1% ^ ^ X , (x - 1) 3 ^ (x + 1) t f ^ W T t , ^ ^ltj'+)d 3, 6 3 ^ 4 f l (a + b) ^ TO ? (a) 3 (b) 2 (c) 1 (d) - 1

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**117. [UNKNOWN] unknown**

44. If the average of 9 consecutive positive integers is 55, then what is the largest integer ? (a) 57 (b) 58 (c) 59 (d) 60

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**118. [UNKNOWN] unknown**

44. ^ 9 shWM ^HlrHch ^ 3 ^ 55 t , ^ W t ? (a) 57 (b) 58 (c) 59 (d) 60

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**119. [UNKNOWN] unknown**

45. The average of the ages of 15 students in a class is 19 years. When 5 new students are admitted to the class, the average age of the class becomes 18-5 years. What is the average age of the 5 newly admitted students ? (a) 17 years (b) 17\*5 years (c) 18 years (d) 18-5 years SPSS-T-XHI ( 1 2 - A ) ===== PAGE 13 =====

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**120. [UNKNOWN] unknown**

45. "QISR % 15 ^ ^ 3TT5 ^ 3 ff^ 19 ^ | ^ W 4 5 ^ ^ ^ 31% fen W T t , cfr ^ 3 t o 3TT5 18-5 ^ ^ t l 3 rf^ ^ 5 ^ ^ 5 T ^ ^ 3 ^ ^ 3TT5 W t ? (a) 1 7 ^ (b) 17-5 ^ (c) 18 ^ (d) 1 8 -5 ^ SPSS-T-XHI ( 1 3 - A ) ===== PAGE 14 =====

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**121. [UNKNOWN] unknown**

46. A»man can row at a speed of x km/hr in still water. If in a stream which is flowing at a speed of y km/hr it takes him z hours to row to a place and back, then what is the distance between the two places ? (a) (b) (c) (d) z ( x ^ - y ^ ) 2y z (x^ - y^ ) 2x (x 2 -y 2 ) 2zx X

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**122. [UNKNOWN] unknown**

46. ^ ^ t \ ^ m\ ( ^ ) , ^ y f ^ / " ^ ^TTT^ t te ft ^ ^FSfR ^ TO t ^ 3 ^ ^TW 3TT^ ( q ^ ) n ^ ^ z ^ W(^ f , ci^ ^ ^2n^ % #ET ^ ^ W t ? (a) (b) (c) (d) z (x^ - y^ ) 2y z (x^ - y^ ) 2x (x ^ -y ^ ) 2zx Z(x2-y2)

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**123. [UNKNOWN] unknown**

47. A water tank has been fitted with two taps P and Q and a drain pipe R. Taps P and Q fill at the rate of 12 litres per minute and 10 litres per minute respectively. Consider the following statements S1, S2 and S3 :

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**124. [UNKNOWN] unknown**

47. ^ ^ T T ^ ^ ^ ^ P 3 f f T Q 3 ^ T T ^ t e m l ^ P 3 ^ Q ^ ^ ^ ^ W ^ : 12 1 W 3 ^ 10 3rf^ f r o ^ ^ ^ f l ^ ^ ^ [HHidP^d ^2R[ s i, S2 3 ^ S3 ^ fe n r : S l : ^ R, 6 ^ 1 w ^ ^ ^ ^ fn^lf^d l

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**125. [UNKNOWN] unknown**

49. A car has an average speed of 60 km per hour while going fi'om Delhi to Agra and has an average speed of y km per hour while returning to Delhi fi'om Agra (by travelling the same distance). If the average speed of the car for the whole journey is 48 km per hour, then what is the value of y ? (a) 30 km per hour (b) 35 km per hour (c) 40 km per hour (d) 45 km per hour An article is sold at a profit of 32%. If the cost price is increased by 20% and the sale price remains the same, then the profit percentage becomes (a) 10% (b) 12% (c) 15% (d) 20% 1 4 - A ===== PAGE 15 =====

**126. [UNKNOWN] unknown**

49.  $3TFTTT \wedge \wedge \wedge 60 f \wedge 3ff \wedge t 3Tk 3TFRT \wedge \wedge FRT W T 353 \wedge 3 ff \wedge -n \wedge y 3rf \wedge TO | cPT \wedge \wedge ) I \wedge \wedge \wedge 3 \wedge ? TcT Tlf \wedge 48 5 [f \wedge T O T \wedge , cfr y \wedge T O \wedge (a) 30 f \wedge 'R I \wedge T O (b) 35 \% Tft3rf \wedge TO (c) 40 \% ift3l1 \% TO (d) 45 f e f t \wedge T O \wedge \wedge 32 \% \wedge \wedge W r t I \wedge eTFM \wedge 20 \% \wedge W T t 3 \wedge \wedge W T t , \wedge \wedge \wedge yfdi \wedge lddl W \wedge t ? (a) 10 \% (b) 12 \% (c) 15 \% (d) 20 \% SPSS-T-XHI ( 1 5 - A ) ===== PAGE 16 =====$

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**127. [UNKNOWN] unknown**

50. A, B, C, D and E start a partnership firm. Capital contributed by A is three times that contributed by D. E contributes half of A's contribution, B contributes one-third of E's contribution and C contributes two-third of A's contribution. If the difference between the combined shares of A, D and E and the combined shares of B and C in the total profit of the firm is  $\wedge 13,500$ , what is the combined share of B, C and E ? (The shares are supposed to be proportional to the contributions) (a)  $r 13,500$  (b)  $\wedge 18,000$  (c)  $?= 19,750$  (d)  $\wedge 20,250$

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**128. [UNKNOWN] unknown**

50. A, B, C, D  $3 \wedge E \wedge \wedge \wedge f I A D \wedge \wedge \wedge cfr \wedge t I E \wedge 3TWT, A \% 3TWT \wedge 3Tmr t, B \wedge 37WT, E \% 3TWT \wedge t, C \wedge 3TW \wedge, A \% 3 iW T \wedge t I \wedge A, D 3 \wedge E \% W B 3 \wedge C \% \wedge T5 \wedge \wedge 3T \wedge \wedge 13,500 t, \wedge B, c 3 \wedge ( E \wedge f | wr t ? (f | \#, 3Tw ff \% w n g \wedge \wedge \wedge \wedge f ) (a) \wedge 13,500 (b) \wedge 18,000$

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**129. [UNKNOWN] unknown**

51 : Pipe R drains out at the rate of 6 litres per minute.

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**130. [UNKNOWN] unknown**

51. A Pie Chart is drawn for the following data : Sector Percentage Agriculture and Rural Development 12\*9 Irrigation 12-5 Energy 27-2 Industry and Minerals 15-4 Transport and Communication 15-9 Social Services 16-1 What is the angle (approximately) subtended by the Social Services Sector at the centre of the circle ? (a) (b) (c) (d)  $45 \wedge 46' 58 = 98'$

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**131. [UNKNOWN] unknown**

51.  $IMeTfecT 3T T \wedge \% \wedge rTte ( \wedge \wedge ) sHNI \wedge FTT \wedge : iw jrfwmm 3i1 < yi41ui f \wedge chm 12-9 ly-ciil 12-5 'dD'Jii 27-2 15-4 15-9 161 (a) 45 = (b) 46 = (c) 58 = (d) 98 = SPSS-T-XHI$

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**132. [UNKNOWN] unknown**

52 : If both the taps and the drain pipe are opened simultaneously, then the tank is filled in 5 hours 45 minutes.

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**133. [UNKNOWN] unknown**

52 :  $\wedge \wedge \wedge 3fk \wedge \wedge \wedge 2T \wedge \wedge W T I, \wedge 5 \wedge 45 f e z \wedge Z \wedge I$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**134. [UNKNOWN] unknown**

52.  $\wedge ws \wedge n \wedge rf \wedge 101 3 \wedge \wedge 3 \wedge Tpir \wedge 8 1 I \wedge Ws \wedge TR f ? (a) 15, 5 (b) 12,8 (c) 16,4 (d) 18,2 (a) 10-0 (b) 10-5 (0 11-0 (d) 11-5$

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**135. [UNKNOWN] unknown**

53 : Pipe R drains out (fully) the filled tank in 15 hours 20 minutes. To know what is the capacity of the tank, which one

of the following is correct ? (a) S2 is only sufficient (b) S1, S2 and S3 are necessary (c) Any two out of S1, S2 and S3 are sufficient (d) None of the above

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**136. [UNKNOWN] unknown**

53.  $\frac{R}{15} = \frac{20}{m}$  and  $\frac{m}{TR} = \frac{f}{\pi h^2}$  ? (a)  $\frac{S}{2} = \frac{7}{3}$  (b)  $S_1, S_2, S_3$  (c)  $S_1, S_2, S_3$  (d)  $3q$

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**137. [UNKNOWN] unknown**

53. The arithmetic mean of two numbers is 10 and their geometric mean is 8. What are the two numbers ? (a) 15, 5 (b) 12, 8 (c) 16, 4 (d) 18, 2  
The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10.5 and the arithmetic mean of the last 6 observations is 11.5. What is the sixth observation ? (a) 10 (b) 10.5 (c) 11 (d) 11.5  
What is  $\sin^2 \theta - \cos^2 \theta$  equal to for any real number  $\theta$  ? (a) 1 (b)

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**138. [UNKNOWN] unknown**

53.  $\frac{11}{m} = \frac{1}{n^4}$  (a)  $\frac{1}{m} = \frac{1}{n^4}$  (b)  $\frac{1}{m} = \frac{1}{n^4}$  (c)  $\frac{1}{m} = \frac{1}{n^4}$  (d)  $\frac{1}{m} = \frac{1}{n^4}$

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**139. [UNKNOWN] unknown**

54.  $\frac{f}{RT} = \frac{W}{n}$  ? (a) 1 (b)

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**140. [UNKNOWN] unknown**

55. What is  $\cot 1^\circ \cot 23^\circ \cot 45^\circ \cot 67^\circ \cot 89^\circ$  equal to ? (a) 0 (b) 1 (c) 2 (d) 3

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/QP-CDSI18-Math.pdf

**141. [UNKNOWN] unknown**

55.  $\cot 1^\circ \cot 23^\circ \cot 45^\circ \cot 67^\circ \cot 89^\circ$  ? (a) 0 (b) 1 (c) -

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**142. [UNKNOWN] unknown**

56. What angle does the hour hand, of a clock describe in 10 minutes of time ? (a)  $1^\circ$  (b)  $5^\circ$  (c)  $6^\circ$  (d)  $10^\circ$   
(a) H - h units  
Consider the following statements : (b) H - 2h units

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**143. [UNKNOWN] unknown**

56.  $\frac{W}{T} = \frac{10}{e}$  ? (a)  $1^\circ$  (b)  $5^\circ$  (c)  $6^\circ$  (d)  $10^\circ$   
(a) H - h units (b) H - 2h units  
(c) H - 3h units

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**144. [UNKNOWN] unknown**

58. Each side of a square subtends an angle of  $60^\circ$  at the tip of a tower of height h metres standing at the centre of the square. If l is the length of each side of the square, then what is  $h^2$  equal to ? (a)  $\frac{2}{2}$  (b) L (c)  $\frac{3}{2}$  (d)  $\frac{2}{2}$

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**145. [UNKNOWN] unknown**

58.  $\frac{h}{T} = \frac{Z}{R} = \frac{M}{60}$  ? (a)  $\frac{2}{2}$  (b)  $\frac{3}{2}$  (c)  $\frac{3}{2}$  (d)  $\frac{3}{2}$

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**146. [UNKNOWN] unknown**

59. From a height of  $h$  units, a man observes the angle of elevation as  $a$  and angle of depression as  $p$  of the top and the bottom respectively of a tower of height  $H$  ( $> 4h$ ). To what further height should he climb so that the values of angle of elevation and angle of depression get interchanged for the top and bottom of the tower ? SPSS-T-XHI ( 1 8 - A ) =====  
PAGE 19 =====

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**147. [UNKNOWN] unknown**

59.  $h \wedge \wedge \wedge \text{scfT t \% H ( > 4 h) } \wedge \wedge \text{3fk } \wedge \wedge \text{SFIT9T: } \wedge \text{a } 3 \wedge \text{3T}^{\wedge} \text{RTCR } \wedge \text{p t l } \wedge \text{3TT}^{\wedge} 3 \wedge \text{feRT } \wedge \text{cTlf } \% \wedge \text{3fk cT}^{\wedge} \% \wedge 3 \wedge \text{37}^{\wedge} \text{WT } \wedge \% \text{ SPSS-T-XHI ( 1 9 - A ) ===== PAGE 20 =====}$

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**148. [UNKNOWN] unknown**

60. If  $\sec X \operatorname{cosec} x = 2$ , then (a) 1 only  $\tan X + \cot X$  equal to ? (a) 2 (b) 2 only (b)  $2n+1$  (c) Both 1 and 2 (c)  $2^{\wedge}$  (d) Neither 1 nor 2 (d)  $2n-1$

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**149. [UNKNOWN] unknown**

61. If  $\cos X + \cos^{\wedge} X = 1$ , then what is  $\sin^{\wedge} X + \sin^{\wedge} x$  equal to ? (a) 1 (b) 1-5 (c) 2 (d) 3

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**150. [UNKNOWN] unknown**

61.  $\cos X + \cos^{\wedge} X = 1 \sin^{\wedge} x + \sin^{\wedge} x \text{ sm i l ? (a) (b) (c) (d) 1 1-5 2 3}$

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**151. [UNKNOWN] unknown**

62. If  $\sin A + \cos A = p$  and  $\sin^{\wedge} A + \cos^{\wedge} A = q$ , then which one of the following is correct ? (a)  $-3p + q = 0$  (b)  $q^{\wedge} - 3q + 2p = 0$  (c)  $p^{\wedge} - 3p + 2q = 0$  (d)  $p^{\wedge} + 3p + 2q = 0$

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**152. [UNKNOWN] unknown**

62.  $\sin A + \cos A = p 3^{\wedge} \sin^{\wedge} A + \cos^{\wedge} A = q \wedge \wedge \wedge \text{t ? (a) } -3p + q = 0 \text{ (b)}$

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**153. [UNKNOWN] unknown**

63. If  $x = \sec^{\wedge} 0 - \tan 0 \sec^{\wedge} 0 + \tan 0$  following is correct ? then which one of the (a)  $-\infty < x < 3$  (b)  $X < 3$  (c)  $-3 < X < -1$  (d)  $i < x < 3$

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**154. [UNKNOWN] unknown**

63.  $\wedge X = \wedge \wedge \sec 0 + \tan 0 \text{ (a) } -\infty < x < 3 \text{ O - , } 3 \text{ (c) } -3 < X < -\infty \text{ U (d) } \wedge S x < 3 \text{ u}$

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**155. [UNKNOWN] unknown**

64. ABC is a right angled triangle with base BC and height AB. The hypotenuse AC is four times the length of the perpendicular drawn to it from the opposite vertex. What is  $\tan C$  68. equal to ? (a)

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**156. [UNKNOWN] unknown**

64.  $ABC \wedge \text{I } 3 \text{mK BC } 3 \text{fk AB } \text{I } \wedge AC \wedge \wedge \wedge \text{m } \wedge \wedge \wedge \wedge \text{I } \tan C \text{ t ? (a)}$

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**157. [UNKNOWN] unknown**

67. ABC is a triangle right angled at C with  $BC = a$  and  $AC = b$ . If  $p$  is the length of the perpendicular from C on AB, then which one of the following is correct ? (a)  $a^2 b^2 = p^2 (a^2 + b^2)$  (b)  $a^2 b^2 = p^2 (b^2 - a^2)$  (c)  $2a^2 b^2 = p^2 (a^2 + b^2)$  (d)  $a^2 b^2 = 2p^2 (a^2 + b^2)$  The radius and slant height of a right circular cone are 5 cm and 13 cm respectively. What is the volume of the cone ? (a)  $100\pi \text{ cm}^3$  (b)  $5071 \text{ cm}^3$  (c)  $6571 \text{ cm}^3$  (d)  $169\pi \text{ cm}^3$  Two equal circular regions of greatest possible area are cut off from a given circular sheet of area A. What is the remaining area of the sheet ? (a)  $A/2$  (b)  $A/3$  (c)  $3A/5$  (d)  $2A/5$  If the ratio of the radius of the base of a right circular cone to its slant height is 1 : 3, what is the ratio of the total surface area to the curved surface area ? (a) 5 : 3 (b) 3 : 1 (c) 4 : 1 (d) 4 : 3 ===== PAGE 21 =====

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**158. [UNKNOWN] unknown**

68. ABC is a triangle right angled at C with  $BC = a$  and  $AC = b$ . If  $p$  is the length of the perpendicular from C on AB, then which one of the following is correct ? (a)  $a^2 b^2 = p^2 (a^2 + b^2)$  (b)  $a^2 b^2 = p^2 (b^2 - a^2)$  (c)  $2a^2 b^2 = p^2 (a^2 + b^2)$  (d)  $a^2 b^2 = 2p^2 (a^2 + b^2)$  The radius and slant height of a right circular cone are 5 cm and 13 cm respectively. What is the volume of the cone ? (a)  $100\pi \text{ cm}^3$  (b)  $5071 \text{ cm}^3$  (c)  $6571 \text{ cm}^3$  (d)  $169\pi \text{ cm}^3$  Two equal circular regions of greatest possible area are cut off from a given circular sheet of area A. What is the remaining area of the sheet ? (a)  $A/2$  (b)  $A/3$  (c)  $3A/5$  (d)  $2A/5$  If the ratio of the radius of the base of a right circular cone to its slant height is 1 : 3, what is the ratio of the total surface area to the curved surface area ? (a) 5 : 3 (b) 3 : 1 (c) 4 : 1 (d) 4 : 3 ===== PAGE 22 =====

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**159. [UNKNOWN] unknown**

70. A right circular cone is sliced into a smaller cone and a frustum of a cone by a plane perpendicular to its axis. The volume of the smaller cone and the frustum of the cone are in the ratio 64 : 61. Then their curved surface areas are in the ratio (a) 4 : 1 (b)

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**160. [UNKNOWN] unknown**

70. A right circular cone is sliced into a smaller cone and a frustum of a cone by a plane perpendicular to its axis. The volume of the smaller cone and the frustum of the cone are in the ratio 64 : 61. Then their curved surface areas are in the ratio (a) 4 : 1 (b)

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**161. [UNKNOWN] unknown**

71. A region of area A bounded by a circle C is divided into n regions, each of area  $A/n$ , by drawing circles of radii  $r_1, r_2, \dots, r_{n-1}$  such that  $r_1 < r_2 < \dots < r_{n-1}$  concentric with the circle C. If  $p_m = +1$  where  $m = 1, 2, 3, \dots, (n-2)$ , then which one of the following is correct ? (a) (0) p increases as m increases p decreases as m increases p remains constant as m increases (d) p increases for some values of m as m increases and then decreases thereafter

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**162. [UNKNOWN] unknown**

71. A region of area A bounded by a circle C is divided into n regions, each of area  $A/n$ , by drawing circles of radii  $r_1, r_2, \dots, r_{n-1}$  such that  $r_1 < r_2 < \dots < r_{n-1}$  concentric with the circle C. If  $p_m = +1$  where  $m = 1, 2, 3, \dots, (n-2)$ , then which one of the following is correct ? (a) (0) p increases as m increases p decreases as m increases p remains constant as m increases (d) p increases for some values of m as m increases and then decreases thereafter

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**163. [UNKNOWN] unknown**

75. What is the volume of a cone of maximum volume cut out from a cube of edge 2a such that their bases are on the same plane ? (a)  $\frac{2\sqrt{3}}{3} a^3$  (b)  $\frac{2\sqrt{3}}{3} a^3$  (c)  $\frac{2\sqrt{3}}{3} a^3$  (d)  $\frac{2\sqrt{3}}{3} a^3$  The radii of two circles are 4\*5 cm and 3-5 cm respectively. The distance between the centres of the circles is 10 cm. What is the length of the transverse common tangent ? (a) 4 cm (b) 5 cm (c) 6 cm (d) 7 cm There are as many square centimetres in the surface area of a sphere as there are cubic centimetres in its volume. What is the radius of the sphere ? (a) 4 cm (b) 3 cm (c) 2 cm (d) 1 cm The length of a line segment AB is 2 cm. It is divided into two parts at a point C such that  $AC^2 = AB \times CB$ . What is the length of CB ? (a) 3 Vs cm (b)

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**164. [UNKNOWN] unknown**

75.  $2a^4 - 5cm^3 + 3 - 5cm^4 + 10cm^3 + 3^3 + Wrf^5[ST^w]$  ? (a)  $Tia^4$  (b) (c) (d)  $na^2na^3Sna^4$   
^4-5 cm 3 ^3-5 cm f l ^4 10 cm t l 3 ^3 ^W r f ^5[ST ^w | ? (a) 4 cm (b) 5 cm (c) 6 cm (d) 7 cm f 3T R cR ^4 f l ^  
^ W t ? (a) 4 cm (b) 3 cm (c) 2 cm (d) 1 cm A B ^2 cm t l % C ^4 m t ^4 ^FPC^t 1% l ^ C B ^ W t ? AC2 = AB X CB1 (a)  
3 Vs cm (b)

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**165. [UNKNOWN] unknown**

76. The locus of the mid-points of the radii of length 16 cm of a circle is (a) A concentric circle of radius 8 cm A concentric circle of radius 16 cm The diameter of the circle

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**166. [UNKNOWN] unknown**

76.  $16cmCl'cjl \wedge \wedge \wedge Hrr^nsft \% Tf^f.3fr^t$  (a)  $8cm^4$  (b)  $16cm^4$  (c)  $o^4TRT$  (d)  $^4WT^4$

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**167. [UNKNOWN] unknown**

77. The curved surface area of a right circular cone is  $1.76m^2$  and its base diameter is 140 cm. What is the height of the cone ? (a) 10 cm (b)  $10\sqrt{2}$  cm (c)  $20\sqrt{2}$  cm (d)  $10\sqrt{15}$  cm

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**168. [UNKNOWN] unknown**

77.  $XT^4 + 1.76m^2 + 3^4mTR^4 140cm^2 + 1^4Wt^4$  ? (a) 10 cm (b)  $10\sqrt{2}$  cm (c)  $20\sqrt{2}$  cm (d)  $10\sqrt{15}$  cm

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**169. [UNKNOWN] unknown**

78. Consider the following statements :

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**170. [UNKNOWN] unknown**

79. (b) (c) (d) A straight line passing through the centre of the circle

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**171. [UNKNOWN] unknown**

79.  $^4yricTE^4 + 3T^4 + ^4^4^4 \%q2l^4$  (a) ^4 (b)  $x^4?^4$  (c)  $Wf^4 + 3T^4 + TT^4 + nr^4$  (d)  $ciM^4chiTt^4 + ^4$

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**172. [UNKNOWN] unknown**

80. Consider the following statements : Two triangles are said to be congruent, if

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**173. [UNKNOWN] unknown**

80.  $fnMf^4Rgd^4 ; ^4il^4TTFR^4 + l^4, ^4$

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**174. [UNKNOWN] unknown**

81: 64 In a room whose floor is a square of side 10 m, an equilateral triangular table of side 2 m is placed. Four book-shelves of size 4 m x 1 m x 9 m are also placed in the room. If half of the rest of the area in the room is to be carpeted at the rate of ^100 per square metre, what is the cost of carpeting (approximately) ? (a) 7,600 (b) 5,635 (c) 4,113 (d) 3,200

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**175. [UNKNOWN] unknown**

81. Given that the angles of a polygon are all equal and each angle is a right angle. Statement-1: The polygon has exactly four sides. Statement-2: The sum of the angles of a polygon having  $n$  sides is  $(3n - 8)$  right angles. Which one of the following is correct in respect of the above statements ? (a) Both Statement-1 and Statement-2 are true and Statement-2 is the correct explanation of Statement-1 (b) Both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of Statement-1 (c) Statement-1 is true but Statement-2 is false (d) Statement-1 is false but Statement-2 is true

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**176. [UNKNOWN] unknown**

81.  $f^{\wedge} W t 1^{\wedge} \% W^{\wedge} \wedge T R T f 3^{\wedge} : 3 R^{\wedge} \wedge 4 H H c h > i t I^{\wedge} W - i ; ^{\wedge} 5 g ; 5 r^{\wedge} \wedge \wedge f | ^{\wedge} m^{\wedge} - 2 : n^{\wedge} 3 T T^{\wedge} \wedge f r n r n^{\wedge} (3 n - s) t i 3 M^{\wedge} c f d^{\wedge} \wedge \wedge t ?$  (a)  $^{\wedge} 2 R - i 3 f k^{\wedge} S R - 2^{\wedge} \wedge f , W^{\wedge} 2 R - 2 , ^{\wedge} S H - 1^{\wedge} \wedge o z m F T T t$  (b)  $^{\wedge} 2 R - 1 3^{\wedge} \wedge 2 R - 2^{\wedge} \wedge I , ^{\wedge} \wedge - 2 , ^{\wedge} m - 1^{\wedge} \wedge I$  (c)  $^{\wedge} S R - 1^{\wedge} I , ^{\wedge} S R - 2 T T^{\wedge} I$  (d)  $^{\wedge} 8 H - 1 I , ^{\wedge} \wedge - 2^{\wedge} t$

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**177. [UNKNOWN] unknown**

82. If the length of a side of a square is increased by 8 cm, its area increases by 120 square cm. What is the length of a side of the square ? (a) 2.5 cm (b) 3.5 cm (c) 4.5 cm (d) 5.5 cm SPSS-T-XHI ( 2 4 - A ) ===== PAGE 25 =====

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**178. [UNKNOWN] unknown**

82.  $^{\wedge} T ? : ^{\wedge} \wedge \wedge \wedge \wedge e i^{\wedge} 8 c m^{\wedge} \wedge e f t - 3^{\wedge} T^{\wedge} 120 c m^2^{\wedge} \wedge T n n I^{\wedge} \wedge \wedge \wedge \wedge I ?$  (a) 2.5 cm (b) 3.5 cm (c) 4.5 cm (d) 5.5 cm SPSS-T-XHI ( 2 5 - A ) ===== PAGE 26 =====

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**179. [UNKNOWN] unknown**

83. What is the largest power of 10 that divides the product  $1 \times 2 \times 3 \times 4 \dots \times 2 \times 3 \times 2 \times 4 \times 2 \times 5$  ? (a) 2 (b) 4 (c) 5 (d) None of the above

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**180. [UNKNOWN] unknown**

83.  $10 W W t^{\wedge} 1 X 2 X 3 X 4 \dots X 23 X 24 X 25^{\wedge} \wedge h i c f l t * ?$  (a) (b) (c) (d)  $3 q^{\wedge} \wedge \wedge$

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**181. [UNKNOWN] unknown**

84. Walls (excluding their roofs and floors) of 5 identical rooms having length, breadth and height 6 m, 4 m and 2.5 m respectively are to be painted. Paints are available only in cans of 1 L and one litre of paint can be used for painting 20 square metres. What is the number of cans required for painting ? (a) 10 (b) 12 (c) 13 (d) 14

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**182. [UNKNOWN] unknown**

84.  $T t^{\wedge} \wedge 5^{\wedge} \wedge c M T^{\wedge} \wedge 6 m , 4 m 3^{\wedge} 2 - 5 m t , ^{\wedge} \wedge f^{\wedge} > M I^{\wedge} H I t I^{\wedge} \wedge 20^{\wedge} \wedge \wedge \wedge \wedge \wedge t 1^{\wedge} \wedge 3 T R^{\wedge} 1 \%^{\wedge} \wedge W t ?$  (a) 10 (b) 12 (c) 13 (d) 14

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**183. [UNKNOWN] unknown**

85.  $^{\wedge} 3 T R m^{\wedge} \wedge T T P f 4 - 5 m 3^{\wedge} \wedge 10 m t , 50 c m^{\wedge} 5^{\wedge} \% c j^{\wedge} l l^{\wedge} h K Z T f^{\wedge} d M I^{\wedge} f I^{\wedge} \% 3 l c^{\wedge} \wedge 20^{\wedge} f 3^{\wedge} ? 1001 I T T F f \% f^{\wedge} Z I^{\wedge} \wedge \wedge T O^{\wedge} ?$  (a) 1,200 (b) 1,100 (c) 1,000 (d) 900  $3 T^{\wedge} \wedge \wedge \wedge \wedge \wedge \wedge \# W I I^{\wedge} \wedge S T T^{\wedge} \% 3 M ? R f^{\wedge} a^{\wedge} q i c T T O I ?$  (a)

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**184. [UNKNOWN] unknown**

86. A rectangular pathway having width 4.5 m and length 10 m will have to be tiled using square tiles of side 50 cm.

Each packet of such tiles contains 20 pieces and costs ₹ 100. What will be the total cost of tiles for the pathway ? (a) 1,200 (b) 1,100 (c) 1,000 (d) 900

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**185. [UNKNOWN] unknown**

87. If the ratio of the circumference of the base of a right circular cone of radius  $r$  to its height is

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**186. [UNKNOWN] unknown**

87.  $\frac{f}{T} \propto r$   $\frac{T}{f} \propto r$   $\frac{f}{T} \propto \frac{1}{r}$   $\frac{T}{f} \propto \frac{1}{r}$   $\frac{f}{T} \propto r^2$   $\frac{T}{f} \propto r^2$   $\frac{f}{T} \propto \frac{1}{r^2}$   $\frac{T}{f} \propto \frac{1}{r^2}$  (a)  $37\pi r^2$  (b)  $0$  (d)  $2\pi r^2 - \frac{y}{47T} + 9$

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**187. [UNKNOWN] unknown**

88. A wire is in the form of a circle of radius 98 cm. A square is formed out of the wire. What is the length of a side of the square ? (Use  $\frac{22}{7} = \frac{22}{7}$ ) (a) (b) (c) (d) 146 cm 152 cm 154 cm 156 cm Consider the following for the next two (02) questions: In a triangle ABC,  $a$ ,  $b$  and  $c$  are the lengths of the sides and  $p$ ,  $q$  and  $r$  are the lengths of its medians.

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**188. [UNKNOWN] unknown**

88.  $\frac{a}{b} = \frac{c}{d}$ ,  $98 \text{ cm} \propto \frac{1}{T}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$  (a) 146 cm (b) 152 cm (c) 154 cm (d) 156 cm  $3P^2 \propto \frac{1}{f}$  (02)  $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$

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**189. [UNKNOWN] unknown**

89. Which one of the following is correct ? (a)  $2(p + q + r) = (a + b + c)$  (b)  $2(p + q + r) > 3(a + b + c)$  (c)  $2(p + q + r) < 3(a + b + c)$  (d)  $11(p + q + r) > 10(a + b + c)$

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**190. [UNKNOWN] unknown**

89.  $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$  (a)  $2(p + q + r) = (a + b + c)$  (b)  $2(p + q + r) > 3(a + b + c)$  (c)  $2(p + q + r) < 3(a + b + c)$  (d)  $11(p + q + r) > 10(a + b + c)$

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**191. [UNKNOWN] unknown**

90. Which one of the following is correct ? (a)  $(a + b + c) < (p + q + r)$  (b)  $3(a + b + c) < 4(p + q + r)$  (c)  $2(a + b + c) > 3(p + q + r)$  (d)  $3(a + b + c) > 4(p + q + r)$  ( 26 - A ) ===== PAGE 27 =====

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**192. [UNKNOWN] unknown**

90.  $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$  ===== PAGE 28 =====

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**193. [UNKNOWN] unknown**

91. What is the area of the largest circular disc cut from a square of side  $\frac{1}{\sqrt{2}}$  units ? (a)  $n$  square units (a)  $12\sqrt{2} \text{ cm}^2$  (b) 1 square unit (b)  $12\sqrt{3} \text{ cm}^2$  (c)  $n^2$  square units (c)  $16\sqrt{2} \text{ cm}^2$  (d) 2 square units (d)  $16\sqrt{3} \text{ cm}^2$

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**194. [UNKNOWN] unknown**

91.  $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$   $\frac{a}{b} = \frac{c}{d}$  (a)  $n^2$  (b) 1 (c)  $7c^2$  (d)  $2$

**195. [UNKNOWN] unknown**

92. The product of the lengths of the diagonals of a square is 50 square units. What is the length of a side of the square ? (a)  $5\sqrt{2}$  units (b) 5 units (c) 10 units (d)  $2\sqrt{d}$  units

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**196. [UNKNOWN] unknown**

92.  $XT^{44}50^{11}$  ? (a)  $5\sqrt{2}^{44}$  (b)  $5^{?4}$  (c)  $10^{44}$  (d)  $2\sqrt{s}$

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**197. [UNKNOWN] unknown**

93. The surface area of a closed cylindrical box is 352 square cm. If its height is 10 cm, then 22 what is its diameter ? (Use  $n = \pi$ ) (a) 4 cm (b) 8 cm (c) 9-12 cm (d) 19-26 cm SPSS-T-XHI ( 2 8 - A ) (a) 70' (b) 65' (c) 60' (d) 55' =====  
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**198. [UNKNOWN] unknown**

93.  $\blacksquare Q;^{44}dHI^K^{44}352^{44}\blacksquare^{44}(i < t \mid ^{10} \text{ cm } t, ^{44} (a) 4 \text{ cm } (b) 8 \text{ cm } (c) 9-12 \text{ cm } (d) 19-26 \text{ cm } \text{SPSS-T-XHI } (b) (tt - 1) a^{44} (c) (7t - 2) a^{44} (d) 2$

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**199. [UNKNOWN] unknown**

94. A square and an equilateral triangle have the same perimeter. If the diagonal of the square is  $6\sqrt{2}$  cm, then what is the area of the triangle ?

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**200. [UNKNOWN] unknown**

94.  $^{44}3fk^{44}mRhim \mid ^{44}i \mid W f 6\sqrt{2} \text{ cm } t, \%3T^{44}t ? (a) (b) (c) (d) 12\sqrt{2} 12 73 16\sqrt{2} 16 73 \text{ cm}^{44} \text{ cm}^{44} \text{ cm}' \text{ cm}$

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**201. [UNKNOWN] unknown**

95.  $^{44}W t^{44}3T^{44}eT^{44}a'^{44}t 3^{44}^{44}^{44} \% (a) \text{ in } - 1) a^{44}$

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**202. [UNKNOWN] unknown**

96. What is the area of the region bounded internally by a square of side of length 'a' and externally by a circle passing through the four comers of the square ? (a)  $(71: - 1) a^2$  square units  $(7 t-l)a^2$  (b) ----- square units (c)  $\text{in } - 2) a^2$  square units  $(n-2)a^2$  (d) ----- square units In the figure given below, XA and XB are two tangents to a circle. If  $\angle AXB = 50^\circ$  and AC is parallel to XB, then what is  $\angle ACB$  equal to ? A

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**203. [UNKNOWN] unknown**

96.  $X A a f k X B T ? : ^{44}^{44}^{44} f \mid ^{44} \angle AXB = 50^\circ 3^{44} AC, XB \% W cR t, \angle ACB ^{44}TRT t ? X (b) 65^\circ (c) 60^\circ (d) 55^\circ ( 2 9 - A )$   
===== PAGE 30 =====

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**204. [UNKNOWN] unknown**

98. In the figure given below, p, q, r are parallel lines; l and m are two transversals. l m

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**205. [UNKNOWN] unknown**

99. In the figure given below, SPT is a tangent to the circle at P and O is the centre of the circle. If  $\angle QPT = a$ , then what is  $\angle POQ$  equal to ? (a)  $a$  (b)  $2a$  (c)  $90^\circ - a$  (d)  $180^\circ - 2a$

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**206. [UNKNOWN] unknown**

99. In the figure given below, SPT is a tangent to the circle at P and O is the centre of the circle. If  $\angle QPT = a$ , then what is  $\angle POQ$  equal to ? (a)  $a$  (b)  $2a$  (c)  $90^\circ - a$  (d)  $180^\circ - 2a$

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**207. [UNKNOWN] unknown**

100. In the figure given below, two equal chords cut at point P. If  $AB = CD = 10$  cm,  $OC = 13$  cm (O is the centre of the circle) and  $PB = 3$  cm, then what is the length of OP ? (a) 5 cm (b) 6 cm (c) 2 cm (d) 2 cm  
SPSS-T-XHI 3 0 - A ) =====  
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**208. [UNKNOWN] unknown**

100. In the figure given below, two equal chords cut at point P. If  $AB = CD = 10$  cm,  $OC = 13$  cm (O is the centre of the circle) and  $PB = 3$  cm, then what is the length of OP ? (a) 5 cm (b) 6 cm (c) 2 cm (d) 2 cm  
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